Penetration Testing Report: Footprinting & Network Scanning Phases
W2-PM-Final | Cybersecurity | NETWORKWALKS

Intern Information	
Pentester Name (Cybersecurity Professional)	Benisse Villasoto
Program/Batch	B084 -Networkwalks
Date	November 10, 2026
Modules completed	1. W2-PM1 (Multiple Kali Tools) 2. W2-PM5 (Zenmap Scanning)
Client Target	1. Networkwalks (secured written permission already) 2. My own local LAN Network
Permission secured from client?	Yes
Phases covered	• Phase 1: Reconnaissance & Footprinting • Phase 2: Scanning & Network Discovery • Phase 5: Report on Findings

Table of Contents

Footprinting & Scanning Report..... 3
1. Liability Disclaimer..... 3
2. Introduction..... 3
3. Tools Used..... 3
4. Activities Performed..... 4
    4.1. Footprinting & Reconnaissance Using Built-In Kali Linux Tools..... 4
    4.2. Network Scanning with Zenmap..... 5
5. Risk Analysis/Impact..... 5
6. Identified Issue & Resolution..... 6
7. Recommendations..... 7
8. Conclusion..... 7
9. Appendices..... 7
    9.1. Collection of Evidence..... 7
    9.2. Author & Project Information..... 12

Footprinting & Scanning Phases Report

---

1. Liability Disclaimer

Everything described here was done only on equipment I own or on systems where I had written permission in advance. The material is meant for educational and research purposes, so please make sure any use of it stays within the law. The instructor, the author, and Networkwalks are not responsible for how this knowledge is applied, and each person is accountable for their own actions. Misuse can lead to criminal charges, heavy fines, loss of employment, and a permanent criminal record. In most countries, accessing a system without authorization is a crime, even if no damage is done.

2. Introduction

This is my Week 2 report for the ongoing internship at Networkwalks, and it covers two modules. In W2-PM1, I footprinted the networkwalks.com domain using a range of Kali Linux tools. In W2-PM5, I used Zenmap to scan my own local network. Taken together, the two modules show how an attacker typically works, starting with gathering publicly available information and then moving on to finding out which hosts on a network are live. I did the footprinting in Kali Linux and the scanning on a Windows PC with Zenmap installed. Each step includes the exact command I ran, the result it produced, a screenshot as evidence, and a short note on why the finding would matter from an attacker's point of view.

3. Tools Used

In this table, the tools employed in this project, together with a description of each tool, are outlined.

Tools Overview	
Tool	**Purpose**
Kali Linux & Windows	These are the operating systems used to carry out the reconnaissance activities. Kali Linux was used for the footprinting tasks, while Windows hosted the scanning and local network checks.
WHOIS	This tool retrieves public registration details for a domain, such as the owner, registration and expiry dates, and the name servers. It helps identify who controls the domain and how long it has been active.
whatweb	This tool fingerprints the technologies running on a website, including the web server, CMS, plugins, and IP address. The results give a clear picture of the site's software stack and any components that may be outdated or vulnerable.
nslookup	This command queries DNS to resolve a domain name into its IP address. It confirms where the domain is hosted and provides the address needed for later scans.

curl	This command sends a request to the website and displays only the HTTP response headers. These headers can reveal the server type, security settings, and other details the site exposes by default.
wafw00f	This tool checks whether a Web Application Firewall is protecting the website and, if so, which product it is. Knowing this tells an attacker how well defended the site is before any further testing.
dnsrecon	This tool enumerates the DNS records of a domain, including NS, MX, SPF, TXT, and SRV records. The output exposes the domain's mail servers, name servers, and other infrastructure that can be mapped further.
Zenmap (Nmap GUI)	This is the graphical interface for Nmap, used to scan the local subnet for active devices. It lists the live hosts along with their IP addresses and MAC addresses, giving a quick view of what is connected to the network.
Windows CMD	The Windows command prompt was used to look up the local machine's network configuration. It provided the IP and MAC addresses, which could then be compared against the Zenmap scan results.

4. Activities Performed

This section details the activities undertaken, organized into two subsections: Footprinting & Reconnaissance and Network Scanning with Zenmap.

4.1. Footprinting & Reconnaissance Using Built-In Kali Linux Tools

The objective of this exercise is to footprint the live website networkwalks.com, which necessitates a set of tools that can each uncover a different aspect of the target while remaining within the built-in capabilities of Kali Linux. In light of this, six tools are employed, namely *whois, nslookup, dnsrecon, curl, whatweb, and wafw00f*. These are all run within Kali's terminal emulator.

1. ***whois*** is used to determine the domain's registration and ownership details. Upon running the command, it yielded results including NetworkWalks' domain ID, server, registry expiry, and other relevant information.
2. ***nslookup*** is used to resolve its name to the corresponding IP addresses and mail records. After executing this command, it gave NetworkWalks' server name and address (i.e., 192.232.216.135).
3. ***dnsrecon*** is used to enumerate its wider DNS records, including name servers and additional hosts. This command provided NetworkWalks' DNS footprint, as shown in the appendices.
4. ***curl*** is then used to examine the HTTP headers and responses of the web server. After running it, it displayed the headers by default.
5. ***whatweb*** is used to identify the technologies underlying the site. This revealed the site's general software stack, such as its MetaGenerator (WordPress 7.1.4 and WP Download Manager 3.3.76).

6. ***wafw00f*** is used to establish whether a web application firewall is present. This revealed that a firewall is currently protecting the site, namely ModSecurity (SpiderLabs) WAF.

Taken together, the findings from these tools allow for a full profile of the target to be compiled, as presented in the succeeding sections, specifically in the appendices.

4.2. Network Scanning with Zenmap

For the network scanning activity, Zenmap was selected as the primary tool, and it was acquired directly from the official website of its developers to ensure that the installation was legitimate. With the setup complete, the Command Prompt was opened and the ***ipconfig*** command was executed so that the machine's own network configuration could be examined, from which the IP address and the LAN subnet were identified. These two values were subsequently entered into the target field of Zenmap, which allowed the tool to direct its scan toward the entire local network rather than a single device. The scan produced a set of findings that included the total number of hosts active within the subnet, as well as the IP address and MAC address assigned to each of them. Beyond the host details, a topology of the network was also generated to give a visual representation of how the discovered devices are connected.

5. Risk Analysis/Impact

Risk Analysis				
#	**Risk**	**Evidence/Observation**	**Potential Impact**	**Risk Level**
1	Public domain registration data (whois)	Registration details for NetworkWalks were openly retrievable.	Phishing, social engineering or attempts to hijack the domain.	Low to Medium
2	Server name and IP address resolved (nslookup)	The domain mapped directly to a server name and IP address.	Direct targeting of the server, skipping any protection tied to the domain name.	Medium
3	DNS records enumerated (dnsrecon)	The site's DNS records were listed, which can include name servers, mail servers and extra hosts.	A wider attack surface, and exposure of the full host list if zone transfers are allowed.	Medium
4	HTTP headers leaking server	The response headers disclosed information	Attackers matching the server version to known	Medium

	details (curl)	about the web server and platform.	vulnerabilities, plus weaknesses from missing security headers.	
5	Software stack fingerprinted (whatweb)	The site's general software stack was identifiable from the outside.	Searches for exploits aimed at those exact technologies, especially if any are outdated.	Medium to High
6	Firewall detection (wafw00f)	The tool showed whether a firewall protects the site and, if so, which product it is.	Research into known bypass techniques for that product, or direct web attacks if no firewall is present.	Medium
7	Local network exposed by the Zenmap scan	The scan revealed the hosts in my subnet along with their IP and MAC addresses.	An attacker on the same network mapping every device, going after weak ones or impersonating a device by its MAC address.	Medium to High

6. Identified Issue & Resolution

1. Ran the nslookup command and it initially returned the following output:

    - nslookup networkwalks.com
      ;; communications error to 8.8.8.8#53: host unreachable
      ;; communications error to 8.8.8.8#53: host unreachable
      ;; communications error to 8.8.8.8#53: host unreachable
      Server: fd17:625c:f037:2::3
      Address: fd17:625c:f037:2::3#53

- ***Potential root cause:*** Earlier, I inadvertently moved Kali Linux into another folder, which caused it to be inaccessible in VirtualBox. This was subsequently resolved when I removed it and re-added it as a machine. Later on, when I attempted to run the nslookup command in its terminal, it returned with an undesired output. I examined Kali network settings in VirtualBox and found that it was reset into its default settings.
- ***Resolution:*** I configured Kali's network settings from NAT to NAT Network (as per the instructions from Week One), and changed all that is necessary. Then, I re-opened Kali Linux and ran the following commands: ip a, which displayed the network interface and IP address, and sudo dhcpcd eth0, which requests network configuration through DHCP for eth0.
- ***Result:*** After the previous configuration and lease request force, I ran the nslookup command again in the terminal, which returned with the target outcome.

7. Recommendations

Most of the information uncovered by the six tools can be concealed through a small number of sound practices. Maintaining the privacy of the domain owner's details, placing the website behind a proxy to conceal its actual address, and removing outdated DNS entries would leave whois, nslookup and dnsrecon with very little to report. Likewise, removing version identifiers from the server, applying standard security headers and keeping all software updated would limit what curl and whatweb are able to disclose, while a web application firewall would provide a front line of defense that wafw00f may detect but cannot easily bypass. Within the local network, dividing it into smaller segments and monitoring for unexpected scans would prevent a tool such as Zenmap from viewing the entire environment at once, and the saved topology files should be regarded as sensitive and restricted to those with a legitimate need to access them. Above all, scanning activities should be conducted only with proper authorization, and each step should be thoroughly documented so that the work can always be demonstrated to be lawful.

8. Conclusion

The second week of my internship gave me new and essential foundations in reconnaissance and the phases of hacking. To recap, I ran six commands in the Kali Linux terminal, each with its own purpose: ***whois*** gave me NetworkWalks' registration details, ***nslookup*** returned the server name and IP address, and ***dnsrecon*** displayed the site's DNS records. ***curl*** showed the HTTP headers, ***whatweb*** revealed the site's general software stack, and ***wafw00f*** told me whether the site is protected by a firewall and, if so, which one. For the network scanning activity, I installed Zenmap from its official website. I then ran ***ipconfig*** in the Command Prompt to get my IP address and LAN subnet, and entered them into Zenmap's target field. The scan showed how many hosts were on my subnet, along with their IP and MAC addresses, which I presented in a table in the previous section. I also generated a topology of my network and saved it as both a PNG and a PDF file.

Overall, these activities showed me how important it is to look at things from an attacker's perspective. Doing so helps me understand their possible motives and moves, and it makes it easier to prevent cyberattacks, as long as everything is done within legal boundaries. They also reminded me how important documentation is. It is useful for future reference, and it also serves as evidence that everything was done lawfully.

9. Appendices

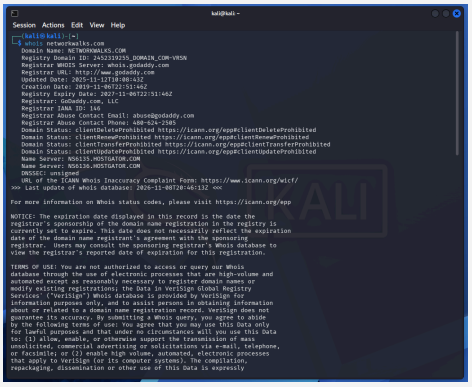
The following section comprises two subsections: the first presents the evidence compiled throughout the project, and the second presents the author and project information.

9.1. Collection of Evidence

In this section, the pieces of evidence compiled during the project are presented in a table. Each command has a corresponding screen capture and textual output obtained from the terminal.

Collection of Evidence

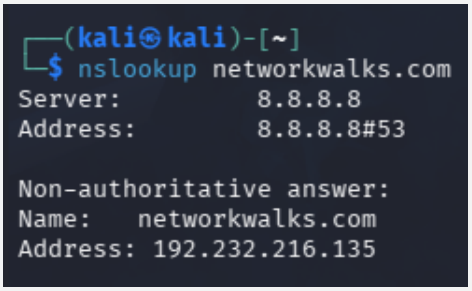
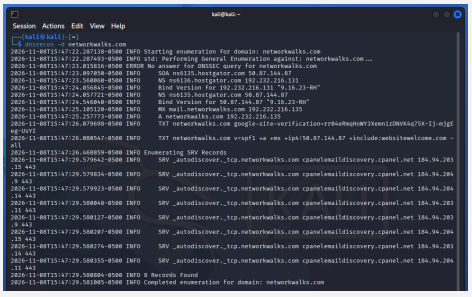
Kali Linux

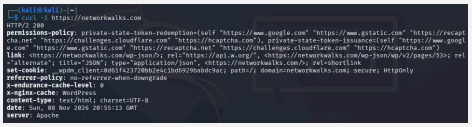
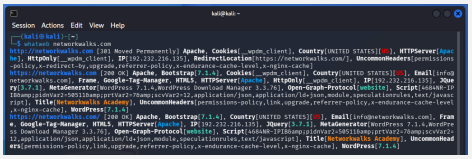
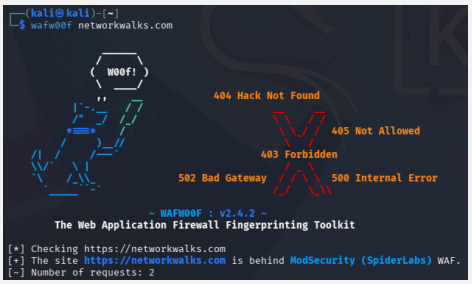
Command	Screen Capture	Complete Textual Output
whois		Domain Name: NETWORKWALKS.COM Registry Domain ID: 2452319255_DOMAIN_COM-VRSN Registrar WHOIS Server: whois.godaddy.com Registrar URL: http://www.godaddy.com Updated Date: 2025-11-12T10:08:43Z Creation Date: 2019-11-06T22:51:46Z Registry Expiry Date: 2027-11-06T22:51:46Z Registrar: GoDaddy.com, LLC Registrar IANA ID: 146 Registrar Abuse Contact Email: abuse@godaddy.com Registrar Abuse Contact Phone: 480-624-2505 Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProhibited Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited Name Server: NS6135.HOSTGATOR.COM Name Server: NS6136.HOSTGATOR.COM DNSSEC: unsigned URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/ >>> Last update of whois database: 2026-11-08T20:40:58Z <<< For more information on Whois status codes, please visit https://icann.org/epp NOTICE: The expiration date displayed in this record is the date the registrar's sponsorship of the domain name registration in the registry is currently set to expire. This date does not necessarily reflect the expiration date of the domain name registrant's agreement with the sponsoring registrar. Users may consult the sponsoring registrar's Whois database to view the registrar's reported date of expiration for this registration. TERMS OF USE: You are not authorized to access or query our Whois database through the use of electronic processes that are high-volume and automated except as reasonably necessary to register domain names or modify existing registrations; the Data in VeriSign Global Registry Services' ("VeriSign") Whois database is provided by VeriSign for information purposes only, and to assist persons in obtaining information about or related to a domain name registration record. VeriSign does not guarantee its accuracy. By submitting a Whois query, you agree to abide by the following terms of use: You agree that you may use this Data only for lawful purposes and that under no circumstances will you use this Data to: (1) allow, enable, or otherwise support the transmission of mass unsolicited, commercial advertising or solicitations via e-mail, telephone, or facsimile; or (2) enable high volume, automated, electronic processes that apply to VeriSign (or its computer systems). The compilation, repackaging, dissemination or other use of this Data is expressly prohibited without the prior written consent of VeriSign. You agree not to use electronic processes that are automated and high-volume to access or query the Whois database except as reasonably necessary to register

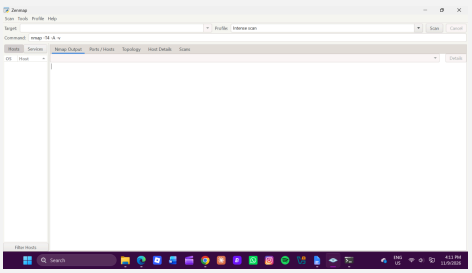
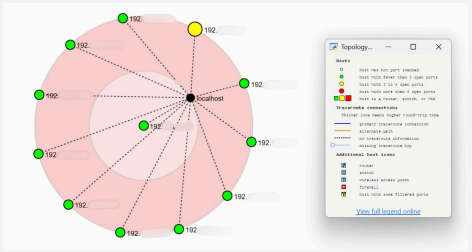
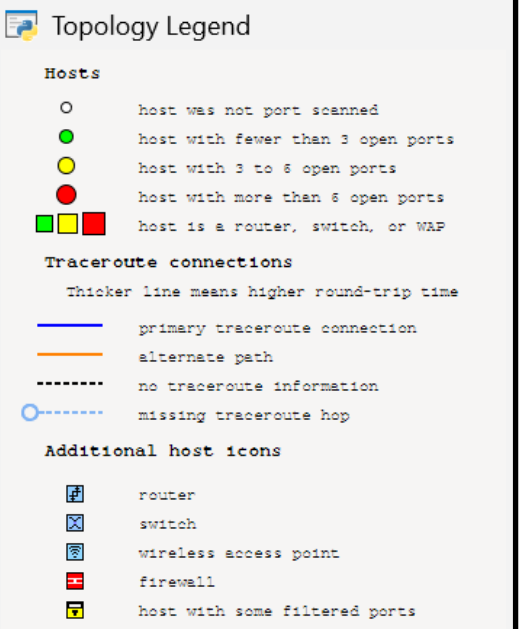
domain names or modify existing registrations. VeriSign reserves the right
to restrict your access to the Whois database in its sole discretion to ensure
operational stability. VeriSign may restrict or terminate your access to the
Whois database for failure to abide by these terms of use. VeriSign reserves the right to modify these terms at any time.

The Registry database contains ONLY .COM, .NET, .EDU domains and
Registrars.
Domain Name: networkwalks.com
Registry Domain ID: 2452319255_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.godaddy.com
Registrar URL: https://www.godaddy.com
Updated Date: 2025-11-12T05:08:41Z
Creation Date: 2019-11-06T17:51:46Z
Registrar Registration Expiration Date: 2027-11-06T17:51:46Z
Registrar: GoDaddy.com, LLC
Registrar IANA ID: 146
Registrar Abuse Contact Email: abuse@godaddy.com
Registrar Abuse Contact Phone: +1.4806242505
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProhibited
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Registry Registrant ID: Not Available From Registry
Registrant Name: Registration Private
Registrant Organization: Domains By Proxy, LLC
Registrant Street: DomainsByProxy.com
Registrant Street: 100 S. Mill Ave, Suite 1600
Registrant City: Tempe
Registrant State/Province: Arizona
Registrant Postal Code: 85281
Registrant Country: US
Registrant Phone: +1.4806242599
Registrant Phone Ext:
Registrant Fax:
Registrant Fax Ext:
Registrant Email: https://www.godaddy.com/whois/results.aspx?domain=networkwalks.com&action=contactDomainOwner
Registry Tech ID: Not Available From Registry
Tech Name: Registration Private
Tech Organization: Domains By Proxy, LLC
Tech Street: DomainsByProxy.com
Tech Street: 100 S. Mill Ave, Suite 1600
Tech City: Tempe
Tech State/Province: Arizona
Tech Postal Code: 85281
Tech Country: US
Tech Phone: +1.4806242599
Tech Phone Ext:
Tech Fax:
Tech Fax Ext:
Tech Email: https://www.godaddy.com/whois/results.aspx?domain=networkwalks.com&action=contactDomainOwner
Name Server: NS6135.HOSTGATOR.COM
Name Server: NS6136.HOSTGATOR.COM
Name Server: NS29.DOMAINCONTROL.COM
Name Server: NS30.DOMAINCONTROL.COM
DNSSEC: unsigned
URL of the ICANN WHOIS Data Problem Reporting System: http://wdprs.internic.net/
>>> Last update of WHOIS database: 2026-11-08T20:41:07Z <<<
For more information on Whois status codes, please visit https://icann.org/epp

TERMS OF USE: The data contained in this registrar's Whois database, while believed by the
registrar to be reliable, is provided "as is" with no guarantee or warranties regarding its
accuracy. This information is provided for the sole purpose of assisting you in obtaining
information about domain name registration records. Any use of this data for any other purpose
is expressly forbidden without the prior written permission of this registrar. By submitting
an inquiry, you agree to these terms and limitations of warranty. In

		particular, you agree not to use this data to allow, enable, or otherwise support the dissemination or collection of this data, in part or in its entirety, for any purpose, such as transmission by e-mail, telephone, postal mail, facsimile or other means of mass unsolicited, commercial advertising or solicitations of any kind, including spam. You further agree not to use this data to enable high volume, automated or robotic electronic processes designed to collect or compile this data for any purpose, including mining this data for your own personal or commercial purposes. Failure to comply with these terms may result in termination of access to the Whois database. These terms may be subject to modification at any time without notice. **NOTICE** This WHOIS server is being retired. Please use our RDAP service instead.
nslookup		Server: 8.8.8.8 Address: 8.8.8.8#53 Non-authoritative answer: Name: networkwalks.com Address: 192.232.216.135
dnsrecon		2026-11-08T15:47:22.287138-0500 INFO Starting enumeration for domain: networkwalks.com 2026-11-08T15:47:22.287493-0500 INFO std: Performing General Enumeration against: networkwalks.com... 2026-11-08T15:47:23.015816-0500 ERROR No answer for DNSSEC query for networkwalks.com 2026-11-08T15:47:23.097050-0500 INFO SOA ns6135.hostgator.com 50.87.144.87 2026-11-08T15:47:23.560060-0500 INFO NS ns6136.hostgator.com 192.232.216.131 2026-11-08T15:47:24.056845-0500 INFO Bind Version for 192.232.216.131 "9.16.23-RH" 2026-11-08T15:47:24.057721-0500 INFO NS ns6135.hostgator.com 50.87.144.87 2026-11-08T15:47:24.546040-0500 INFO Bind Version for 50.87.144.87 "9.16.23-RH" 2026-11-08T15:47:25.105120-0500 INFO MX mail.networkwalks.com 192.232.216.135 2026-11-08T15:47:25.257773-0500 INFO A networkwalks.com 192.232.216.135 2026-11-08T15:47:26.079698-0500 INFO TXT networkwalks.com google-site-verification=rr04eRmqHoWY3XemnizDNVK4q75X-Ij-mjgEeg-UsYI 2026-11-08T15:47:26.080547-0500 INFO TXT networkwalks.com v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com ~all 2026-11-08T15:47:26.468859-0500 INFO Enumerating SRV Records 2026-11-08T15:47:29.579642-0500 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.203.15 443 2026-11-08T15:47:29.579834-0500 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.204.9 443 2026-11-08T15:47:29.579923-0500 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.204.14 443 2026-11-08T15:47:29.580048-0500 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.203.11 443 2026-11-08T15:47:29.580127-0500 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.203.9 443 2026-11-08T15:47:29.580207-0500 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.204.15 443

		2026-11-08T15:47:29.580274-0500 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.203.14 443 2026-11-08T15:47:29.580355-0500 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.204.11 443 2026-11-08T15:47:29.580804-0500 INFO 8 Records Found 2026-11-08T15:47:29.581005-0500 INFO Completed enumeration for domain: networkwalks.com
curl		HTTP/2 200 permissions-policy: private-state-token-redemption=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com"), private-state-token-issuance=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com") link: <https://networkwalks.com/wp-json/>; rel="https://api.w.org/", <https://networkwalks.com/wp-json/wp/v2/pages/53>; rel="alternate"; title="JSON"; type="application/json", <https://networkwalks.com/>; rel=shortlink set-cookie: __wpdm_client=0d61f423720bb2e4c1bd6929babdc9ac; path=/; domain=networkwalks.com; secure; HttpOnly referrer-policy: no-referrer-when-downgrade x-endurance-cache-level: 0 x-nginx-cache: WordPress content-type: text/html; charset=UTF-8 date: Sun, 08 Nov 2026 20:55:13 GMT server: Apache
whatweb		http://networkwalks.com [301 Moved Permanently] Apache, Cookies[__wpdm_client], Country[UNITED STATES][US], HTTPServer[Apache], HttpOnly[__wpdm_client], IP[192.232.216.135], RedirectLocation[https://networkwalks.com/], UncommonHeaders[permissions-policy,x-redirect-by,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache] https://networkwalks.com [200 OK] Apache, Bootstrap[7.1.4], Cookies[__wpdm_client], Country[UNITED STATES][US], Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], HttpOnly[__wpdm_client], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1.4,WordPress Download Manager 3.3.76], Open-Graph-Protocol[website], Script[4684NR-IPIB&pidnVar2=50511&prtVar2=7&scvVar2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1.4] https://networkwalks.com/ [200 OK] Apache, Bootstrap[7.1.4], Country[UNITED STATES][US], Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1.4,WordPress Download Manager 3.3.76], Open-Graph-Protocol[website], Script[4684NR-IPIB&pidnVar2=50511&prtVar2=7&scvVar2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1.4]
wafw00f		404 Hack Not Found 405 Not Allowed 403 Forbidden 502 Bad Gateway 500 Internal Error

		~ WAFW00F : v2.4.2 ~ The Web Application Firewall Fingerprinting Toolkit [*] Checking https://networkwalks.com [+] The site https://networkwalks.com is behind ModSecurity (SpiderLabs) WAF. [~] Number of requests: 2
Zenmap		
GUI		N/A
Topology		

9.2. Author & Project Information

The table below presents the author and project information, together with the author's LinkedIn profile link and the project's GitHub repository link.

Table of Information	
Author	Benisse Villasoto
Internship Batch No.	B084
LinkedIn	*https://www.linkedin.com/in/benisse-villasoto/*

Project	Cybersecurity program at Networkwalks | Week: 02 |
Github Repository	***https://github.com/nisse0903/-NETWORKWALKS-BENISSE-B084-WK2-PM1-PENETRATION-TESTING***